Evaluate 2
LMM安全 2
finetuning 3
MultiModal 1

本文索引

Introduction
A Brief History of Agents
Introduction to Code Agents
Secure Code Execution
Monitoring and Evaluating your Agent
Build a Deep-Research Agent

Building Code Agents with Hugging Face smolagents

2025-09-23 21:40 / 10 / 0 / 0 / 10376 Agent

课程链接

github

Introduction

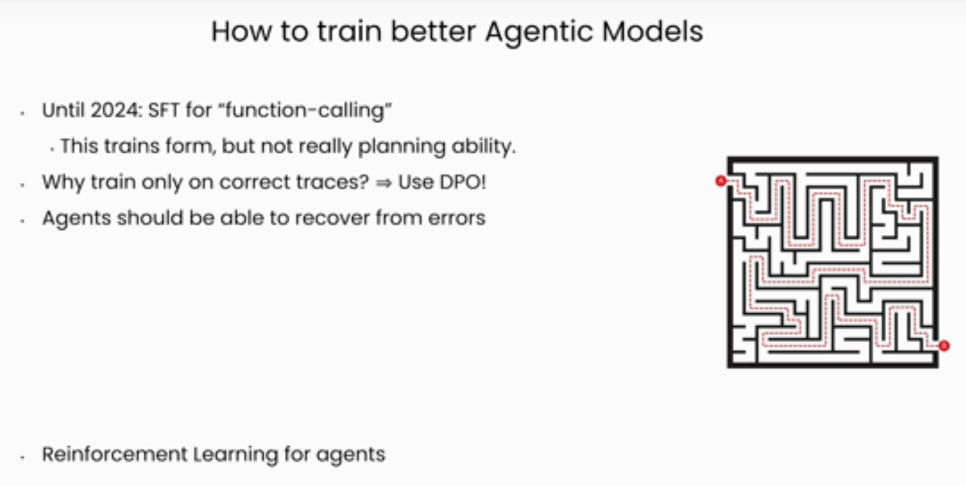
本课程是由huggingface 带来的code agent课程，smolagents是huggingface的框架。本课程讲解了如何安全执行code以及按照ReAct（Thought-Action behavior）的方法搭建相关agent。

A Brief History of Agents

A Range of Agency Levels

Agency Level	Description	How that's called	Example Pattern
☆☆☆	LLM output has no impact on program flow	Simple processor	`process_llm_output(llm_response)`
★☆☆	LLM output determines basic control flow	Router	`if llm_decision(): path_a() else: path_b()`
★★☆	LLM output determines function execution	Tool call	`run_function(llm_chosen_tool, llm_chosen_args)`
★★★	LLM output controls iteration and program continuation	Multi-step Agent	`while llm_should_continue(): execute_next_step()`
★★★	One agentic workflow can start another agentic workflow	Multi-Agent	`if llm_trigger(): execute_agent()`

The Road Towards Increased Agency



Introduction to Code Agents

Secure Code Execution

Threat Vectors

What could be the root cause of malicious code execution?

- **LLM error**
- **Supply chain attack**
- **Prompt injection**

Associated Dangers

- Harm file system
- Steal data
- Create backdoors / install malwares
- Abuse resources: local resources or API services
- Compromise your network

Mitigation: Python Executor with Safeguards

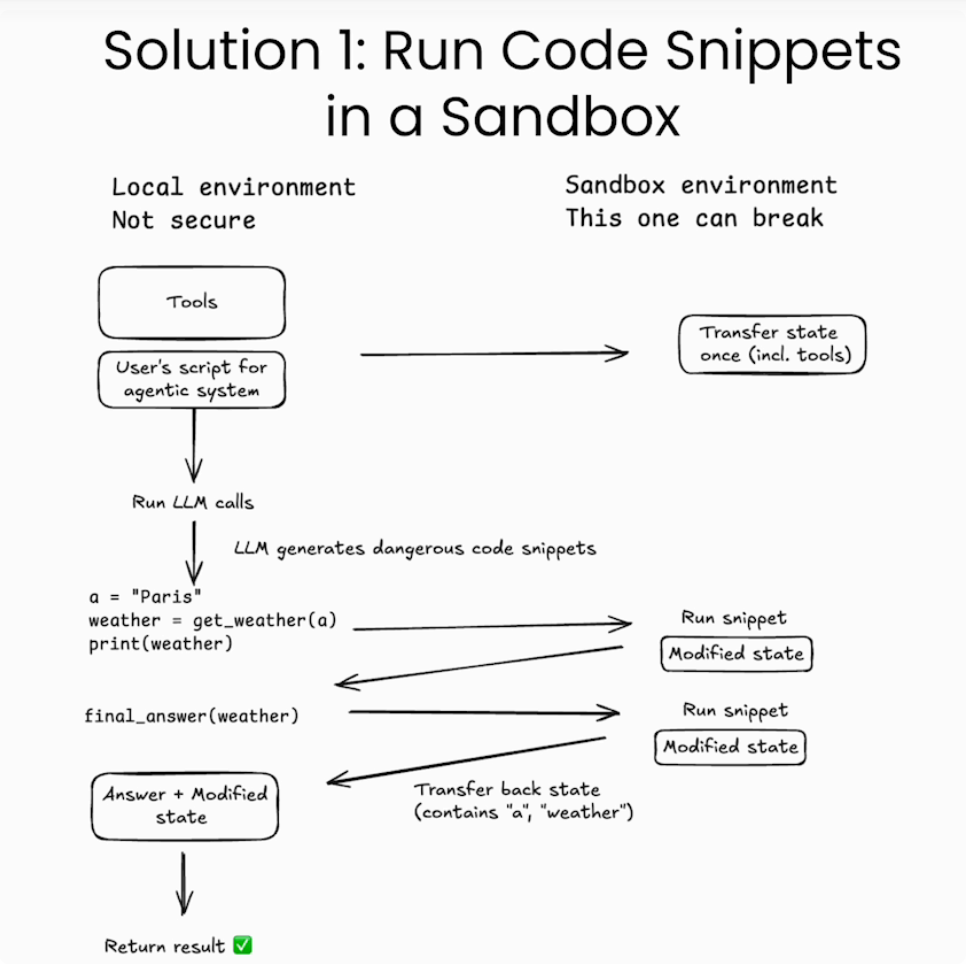
👷 Custom python executor has been re-built from scratch to be more secure.

Built-in safeguards:

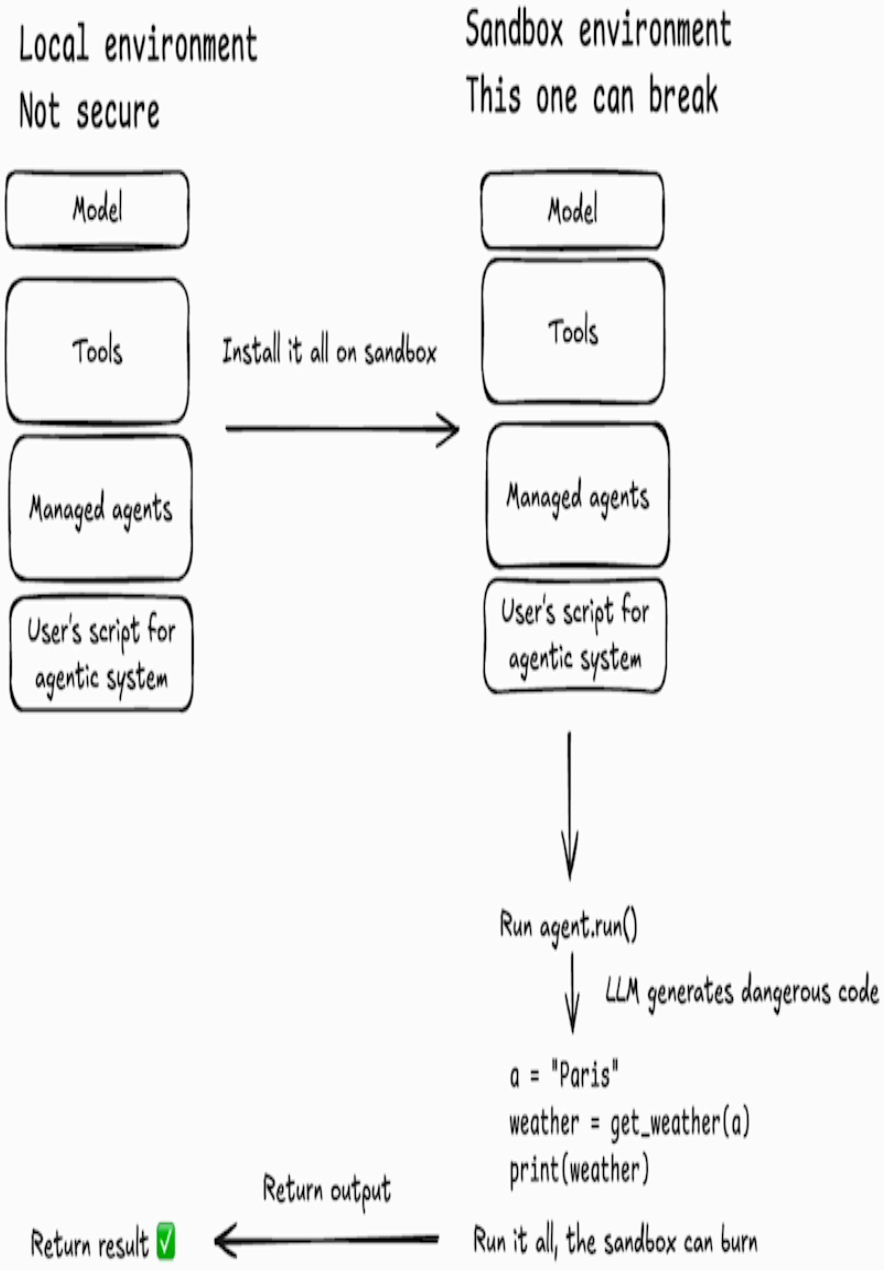
- Any undefined command is ignored
- Import securitization
- Prevent infinite loops

⚠️ These safeguards mitigate danger, but do not fully suppress it!

通过代码安全机制可以限制某些恶意代码增加安全性。


Solution 1: Run Code Snippets in a Sandbox
Local environment
Not secure
Sandbox environment
This one can break
Tools
User's script for agentic system
Transfer state once (incl. tools)
Run LLM calls
LLM generates dangerous code snippets
a = "Paris"
weather = get_weather(a)
print(weather)
Run snippet
Modified state
final_answer(weather)
Run snippet
Modified state
Answer + Modified state
Transfer back state (contains "a", "weather")
Return result ✅

Solution 2: Run the whole Agentic System in a Sandbox



This solution eliminates code execution risk, but requires transferring LLM inference logic to server, possibly API keys

在这里通过PHONIEX平台可以清晰看到Agent运行过程。

Prompt

```
You are an expert assistant who can solve any t
To do so, you have been given access to a list
To solve the task, you must plan forward to pro

At each step, in the 'Thought:' sequence, you s
Then in the 'Code:' sequence, you should write
During each intermediate step, you can use 'pri
These print outputs will then appear in the 'Ob
In the end you have to return a final answer us

Here are a few examples using notional tools:
---
Task: "Generate an image of the oldest person i
```

Build a Deep-Research Agent

在这节为了使得Agnet理解任务更为准确，采用了Multi-Agent。

```
manager_agent.visualize()
```

```
CodeAgent | gpt-4o
├── ✅ Authorized imports: ['geopandas', 'plotly', 'shapely',
'json', 'pandas', 'numpy']
├── 🛠 Tools:
```

Name	Description	Arguments
final_answer	Provides a final answer to the given problem.	answer (`any`): The final answer to the problem

```
└── 🤖 Managed agents:
    └── web_agent | CodeAgent | gpt-4o
        ├── ✅ Authorized imports: []
        ├── 📝 Description: Runs web searches for you.
        └── 🛠 Tools:
```

Name	Description	Arguments
web_search	Searches the web for your query.	query (`string`): Your query
visit_webpage	Visits a webpage at the given url and reads its content as a markdown string. Use this to browse webpages.	url (`string`): The url of the webpage to visit.
final_answer	Provides a final answer to the given problem.	answer (`any`): The final answer to the problem

赞 0

如果觉得文章对你有用，请随意赞赏

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https://halo.mosuyang.org/archives/building-code-agents-with-hugging-face-smolagents

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Administrator	2025-09-23	2025-09-23	CC BY 4.0

Post-training of LLMs

Getting Structured LLM Output

评论